Innovation Centre for Education

Yenepoya Institute of Arts Science Commerce and Management

High Level Design

on

Develop a virtual museumto explore artifacts in 3D

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1. Introduction

The Virtual Museum is a three-dimensional interactive application designed to provide users with an immersive and engaging experience of exploring artifacts digitally. The system is developed using **Unity**, which enables real-time rendering and smooth navigation within a simulated museum environment. Users can move through virtual galleries, observe artifacts in detail, and access additional information such as descriptions, images, or audio explanations. The primary goal of this project is to enhance accessibility to cultural and historical content by allowing users to explore museum exhibits from any location, thereby overcoming physical and geographical limitations.

1.1 Scope of the Document

This document outlines the high-level architecture and overall design of the Virtual Museum system. It describes the major components, system workflows, and data handling mechanisms involved in the application. The scope also includes performance considerations, system interfaces, and non-functional requirements necessary for building a scalable and efficient system.

1.2 Intended Audience

This document is intended for:

- Developers
- Designers
- Project Managers
- Stakeholders

1.3 System Overview

The system is a real-time 3D application that allows users to:

- Navigate a virtual museum
- Interact with artifacts
- View detailed information

The system can operate as a standalone application or integrate with backend services.

2. System Design

The system follows a modular architecture that separates the application into distinct functional components. The frontend is built using Unity, which is responsible for rendering the 3D environment, handling user input, and managing interactions. An optional backend system can be integrated to store and retrieve artifact data through APIs, while a database is used to maintain structured information about artifacts. The overall flow of the system begins with user interaction, which is processed by the application and, if required, communicated to the backend for data retrieval before displaying the results to the user.

2.1. Application Design

The application is divided into multiple modules to ensure proper functionality and maintainability.
Modules:

- Navigation module: Handles movement
- Interaction module: Detects object selection
- UI module: Displays information
- Data module: Manages data

2.2. Process Flow

The system follows a structured workflow starting from user entry to interaction with artifacts. The processensures smooth navigation and information retrieval.
Steps:

1. User launches the application
2. Enters the museum
3. Navigates environment
4. Selects artifact
5. Views information

2.3. Information Flow

The system processes user input and converts it into

actionswithin the virtual environment. Data flows from user interaction to the display system.
Flow:

- User input → Navigation
- Interaction → Data retrieval
- Data → UI display

2.4. Components Design

The system consistsof multiple components that work together to deliver the overall functionality.
Main components:

- 3D environment
- Artifact models
- Camera controller
- UI system
- Data manager

2.5. Key Design Considerations

The design focuses on performance, usability, and scalability to ensure a smooth and efficient system.
Considerations:

- Optimizedmodels
- Smooth navigation
- Scalable design

2.6. API Catalogue

If backend integration is used, APIs will be responsible for handling data requests and responses.
APIs:

- GET /artifacts
- GET /artifact/{id}
- POST /analytics

3. Data Design

The data designfocuses on organizing and managing artifact-related information efficiently so that it can be easily accessed, updated, and displayed within the system. The data is structured to support scalability andquick retrieval during user interaction.
Data includes:

- ArtifactID (unique identifier)
- Name ofthe artifact
- Description (textual details)
- 3D model file path (FBX/OBJ)
- Image or preview file
- Audio/Video description (optional)
- Category (e.g., history, science, art)
- Location within museum (room/section)

3.1 Data Model

The data modeldefines how artifact information is structured and stored within the system.
Key elements:

- Entity:Artifact
- Attributes: ID, name, description, media
- Relationships: Category → Artifacts

3.2 Data Access Mechanism

The system retrieves data either locally or from external sources depending on the design.

Methods:

- Local JSON/XML files
- Database queries
- API-based retrieval

3.3 Data Retention Policies

The system ensures proper storage and maintenance of data over time.

Policies:

- Artifact data stored permanently
- User session data stored temporarily
- Logs stored for analysis

3.4 Data Migration

The systemsupports updates and expansion of data without affecting performance.

Supports:

- Adding new artifacts
- Updating existing data
- Import/export via JSON

4. Interfaces

The system includes both user-facing and external interfaces. The user interface provides controls for navigation and interaction, as well as panels for displaying artifact information. External interfaces include connections to APIs and databases that enable data retrieval and storage. These interfaces ensure smooth communication between different components of the system.

5.State and Session Management

Thesystem maintains the state of the application during runtime by tracking the user's position, current scene, and interactions. This ensures continuity in the user experience and allows the system

to respond appropriately to user actions. Session management helps in maintaining temporary data during the user's interaction with the application.

6. Caching

Caching is used to improve system performance by storing frequently accessed data and preloading nearby assets. This reduces loading times and ensures that the user experiences smooth transitions while navigating the virtual environment. Efficient caching mechanisms are essential for maintaining responsiveness in the application.

7. Non-Functional Requirements

The system must meet several non-functional requirements to ensure quality and reliability. Security measures should be implemented to protect data and prevent unauthorized access. Performance requirements include maintaining a stable frame rate and minimizing latency. The system should also be scalable to support additional content and features. Usability is another key factor, requiring intuitive controls and a user-friendly interface.

8. References

I have used several online platforms such as CGTrader, Sketchfab, and the Unity Asset Store to download and integrate various 3D models, tools, and assets into this project. These resources provided high-quality materials that helped enhance the overall design, functionality, and visual appeal of the application. Additionally, I referred to multiple online tutorials and documentation to better understand the implementation and optimization of these assets within the project.